

BMO RAG CHATBOT - DESIGN BRIEF				
PROJECT PURPOSE	TARGET USERS	PROJECT GOALS	SUCCESS CRITERIA	PROJECT SCOPE
	Internal users who need quick access to research documents, regulatory information, and reports		Ideal satisfaction vs cost trade-off	Core RAG Implementation: Three architecture versions (Basic, Evaluator, Memory)
	CONSTRANTS		High scalability	Production Deployment: Docker + Kubernetes with Minikube support
Enable instant 24/7 assistance with internal documentation	Budget (API & Energy)	Cost Optimization: high performance-cost trade-off	Low latency	Comprehensive Monitoring: Prometheus + Grafana with 15+ metrics
	Latency	Scalability: support of horizontal scaling	ASSUMPTIONS	Quality Assurance: LLM optimization based on context adherence, LLM evaluation, and user feedback
	Regulations	Continuous Improvement: pipeline for continuous improvement based on user feedback	Security Compliance: RAG chatbot meets data protection requirements	
RESOURCES (MATERIAL)			RISKS	
Hosting: Minikube			Hallucination / consistency of answers	
Orchestration: Kubernetes			Security breaches of data by knowledge distillation	
Containerization: Docker			External llm's provider (outage)	
Scaling AI workflow: LangGraph				
Metrics collection: Prometheus				
Visualization: Grafana				
LLM: OpenAI				
Vector database: Chroma				
Web framework: Streamlit				
Storage: EmptyDir				